

Pankaj Kumar Jangid

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SUMMARY

PhD researcher specializing in stochastic modeling, Monte Carlo simulation, and statistical inference with applications to complex dynamical systems. Experienced in large-scale numerical simulations, machine learning methods, and probabilistic modeling using Python and C++. Interested in quantitative research roles involving time-series modeling and stochastic systems in financial markets.

EXPERIENCE

Senior Research Fellow

Aug 2023 - present

- Led theoretical modeling projects on stochastic transport and reaction dynamics resulting in multiple first-author publications in high-impact journals.
- Developed analytical and computational frameworks for transition-path dynamics in non-Markovian systems across rough energy landscapes
- Designed discrete-state stochastic models for nanoscale catalytic processes and temperature-dependent dynamic catalysis
- Built predictive probabilistic models for biomolecular target-search dynamics in structured chromatin environments using stochastic resetting formalisms
- Performed large-scale Monte Carlo simulations of stochastic search and transport processes in heterogeneous environments
- Collaborated with international research groups (Rice University, University of Potsdam) on interdisciplinary stochastic modeling problems

Junior Research Fellow

Aug 2021 - Jul 2023

- Developed stochastic simulation frameworks for chemical and biological dynamical systems using Monte Carlo methods and statistical inference techniques
- Implemented computational pipelines in Python and C++ for trajectory-based simulations of diffusion-controlled processes
- Designed numerical methods for modeling first-passage dynamics and transition-path statistics in stochastic systems
- Performed parameter estimation and validation of probabilistic dynamical models using regression and statistical learning approaches
- Executed large ensemble simulations on high-performance computing (HPC) environments for high-dimensional stochastic systems

EDUCATION

2021 - present PhD (Chemical Physics) at **Indian Institute of Science Education and Research (IISER) Pune**

2016 - 2021 Masters's Degree at **Indian Institute of Science Education and Research (IISER) Mohali**

SKILLS

Programming	Python (NumPy, SciPy, pandas, PyTorch), C++, MATLAB, Mathematica
Statistical & Machine Learning Methods	Regression, Classification, PCA, SVM, Time-series analysis, Neural networks (CNNs, RNNs, VAEs)
Quantitative Modeling	Stochastic processes, Monte Carlo simulation, Non-Markovian dynamics, Brownian dynamics, Statistical inference
Numerical & Computational Methods	Numerical optimization, Scientific computing, High-performance computing (HPC)
Mathematics for Modeling	Probability theory, Statistics, Dynamical systems modeling

PROJECTS

PCA-Based Equity Research Pipeline

[GitHub](#)

Developed a quantitative finance research pipeline for the Nifty stocks using PCA based factor modeling and residual analysis to generate long/short equity signals. The project involved return-matrix construction, rolling latent factor extraction, portfolio signal generation, and out-of-sample performance evaluation using Python.

Generative AI for Molecular Structure Modeling

[GitHub](#)

Developed probabilistic generative models for molecular structure generation using Recurrent Neural Networks(RNN) and Transformer architecture in Python. Implemented stochastic sampling pipelines for high-dimensional molecular configurations. Designed preprocessing and training workflows for sequence-based molecular representations and evaluated generated structures using statistical similarity metrics and validity constraints.

PUBLICATIONS

- **Jangid, P.**, Chaudhury, S. (2026). DNA Target Search in Chromatin Compartments under Stochastic Resetting. *The Journal of Physical Chemistry B*, 130(13), 3766–3775.
- **Jangid, P.**, Chaudhury, S. (2026). Stochastic dynamics of nanoparticle catalysis: a discrete-state perspective. *Materials Horizons*, 13(5), 2152–2172.
- **Jangid, P.**, Metzler, R., Chaudhury, S. (2025). Internal protein motion in a rough model potential. *The Journal of Chemical Physics*, 163(24).
- **Jangid, P.**, Chaudhury, S., Kolomeisky, A. (2025). Effect of Temperature in Dynamic Catalysis. *The Journal of Physical Chemistry C*, 129(35), 15606–15618.
- **Jangid, P.**, Chaudhury, S. (2024). Transition Path Dynamics of Non-Markovian Systems across a Rough Potential Barrier. *The Journal of Physical Chemistry A*, 128(46), 10041–10052.
- **Jangid, P.**, Chaudhury, S., Kolomeisky, A. (2024). Theoretical Understanding of Dynamic Catalysis. *The Journal of Physical Chemistry C*, 128(22), 9077–9089.
- **Jangid, P.**, Punia, B., Chaudhury, S. (2024). Stochastic dynamics of hairballs in single-polymer growth. *Physical Chemistry Chemical Physics*, 26(48), 29749–29758.
- Chaudhury, S., **Jangid, P.**, Kolomeisky, A. B. (2023). Dynamics of chemical reactions on single nanocatalysts with heterogeneous active sites. *The Journal of Chemical Physics*, 158(7).
- **Jangid, P.**, Chaudhury, S. (2023). Exploring the role of heterogeneity in quorum sensing cells: a discrete state stochastic approach. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9), 093501.